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TBD

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Abstract

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Chapter 1.

Introduction

1.1. Motivation

motivation

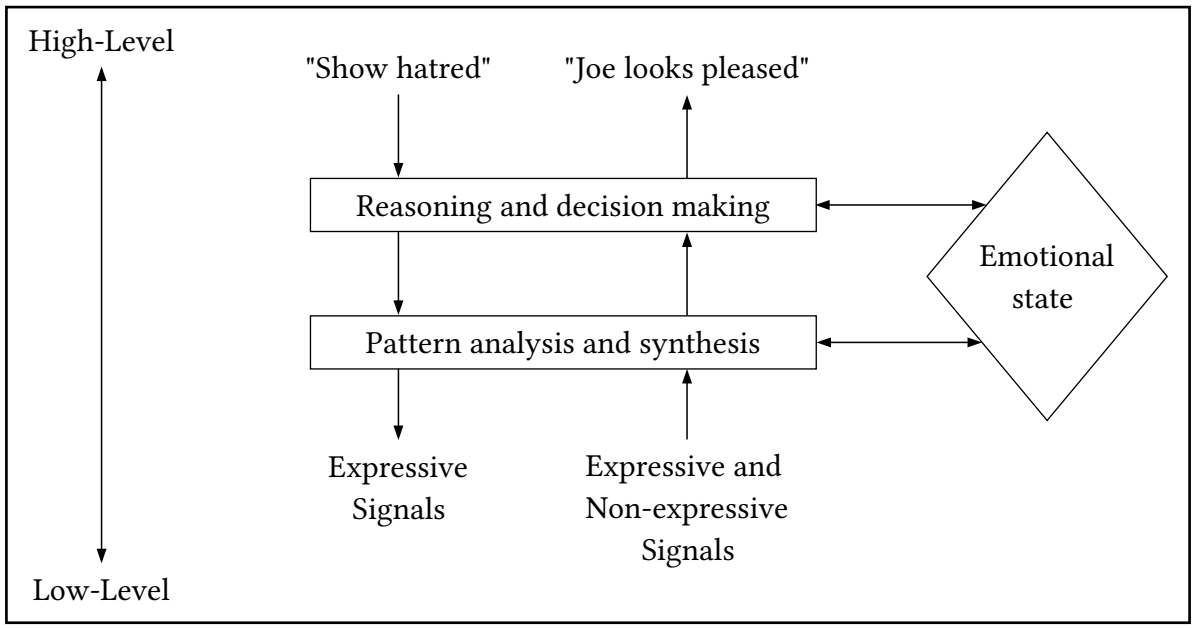


Figure 1.1.: Information flows from both high to low to high in a system that can express, recognize and “have” emotions [1].

1.2. Structure

structure

Chapter 2.

Background

This chapter will give an introduction and overview on different approaches to classify types of emotions with respect to their advantages and disadvantages. Furthermore it will explain the approaches to recognize emotion in the mediums text, video and audio. Based on this common implementation techniques and architectures for Neural Networks are presented.

2.1. Emotion

There is no unambigious definition for the term ‘emotion’.

2.1.1. Emotion Classification

The first frameworks to classify emotions are influenced by Darwin

2.1.2. Emotion in Text

2.1.3. Emotion in Speech

Emotion / Feature	Fear	Anger	Sadness	Hapiness	Disgust
Speech Rate	much faster	slightly faster	slightly slower	faster or slower	very much slower
Pitch Average	very much higher	very much higher	slightly lower	much higher	very much lower
Pitch Range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	normal	higher	lower	higher	lower
Voice Quality	irregular voicing	breathy chest tone	resonant	breathy blaring	grubled chest tone
Pitch Changes	normal	abrupt on stressed syllables	downward inflections	smooth upward inflections	wide downward terminal inflections
Articulation	precise	tense	slurring	normal	normal

Table 2.1.: Features of an audio with correlating emotion [2]

2.1.4. Emotion in Facial Expression

talk about the facs system

2.2. Attention in Deep Learning

Bibliography

[1] R. W. Picard, “Affective computing”. MIT Press, Cambridge, Mass. [u.a.], 1997.

[2] I. R. Murray and J. L. Arnott, “Applying an analysis of acted vocal emotions to improve the simulation of synthetic speech”, *Computer Speech & Language*, vol. 22, no. 2, pp. 107–129, 2008, [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0885230807000393

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Appendix A.

Appendix

Statement of Authorship

I herewith assure that I wrote the present thesis independently, that the thesis has not been partially or fully submitted as graded academic work and that I have used no other means than the ones indicated. I have indicated all parts of the work in which sources are used according to their wording or to their meaning.

I am aware of the fact that violations of copyright can lead to injunctive relief and claims for damages of the author as well as a penalty by the law enforcement agency.

Magdeburg, July 31, 2024

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Signature